

Xutao Gao

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EDUCATION

University of Central Florida

Bachelor of Science (BS)

Computer Science

Orlando, FL

- **GPA:** 4.0, Summa Cum Laude, **Awards:** Provost Scholar, President's Honor Roll
- **Relevant Courses:** Computer Science II, Object-Oriented Programming, System Software, Algorithms for Machine Learning, Database Systems, Senior Design
- **Security Clearance:** Secret (2025 – 2027)

EXPERIENCE

Institute for Simulation & Training at UCF | Full-Stack Engineer

Mar 2025 – Present

- Developed a configurable microchip manufacturing simulation, managing simulation logic, states, and real-time 3D visualization, enabling floor managers to detect operational constraints and optimize workflows.
- Designed and implemented a robust role-based access and approval system for an internal asset management tool, ensuring authorization, workflow integrity, and secure handling of assets from creation to publication.
- Boosted simulation performance 100%+ by implementing an event-driven architecture leveraging lifecycle hooks to reduce calculations and rendering, reducing CPU usage and enabling smooth real-time operation.

PROJECTS | [Github.com/XutaoG](https://github.com/XutaoG)

MeDT Microchip Fabrication Simulator | Simulation WebApp

Sep 2025 – Feb 2026

- Implemented a full-stack, time-based manufacturing simulation platform modeling a complex microchip fabrication environment with 44 machines, 22 processes, and arbitrarily configurable product flows.
- Built an interactive 3D fabrication floor visualization using Angular, NestJS, and WebGL, rendering machine states, worker movement, product lots, and operational metrics with adjustable simulation speed.
- Enabled what-if analysis through configurable products, machines, maintenance schedules, and staffing to expose bottlenecks and inefficiencies.

NSF Digital Marketplace | Full-Stack WebApp

Mar 2025 – Sep 2025

- Designed and implemented an internal full-stack digital marketplace used by 10+ universities and companies to manage and share physical and digital assets.
- Built a role-based access and approval workflow (Owner/Admin/Member) using React, ASP.NET Core, and Azure, supporting asset submission, review, and controlled publication.
- Replaced fragmented email/spreadsheet processes with a centralized platform, streamlining cross-organization asset discovery, requests, and lifecycle management at a large scale.

Online Project Collaboration Platform | Web/Mobile Application

July 2024

- Led frontend development of a platform for student collaborative project work, designing state management to optimize data fetching and support scalable feature expansion.
- Built key features including state management and role-based access control using the MERN stack (web) and Flutter (mobile), following Agile practices with Jira, feature-branch workflow, and code reviews.

CERTIFICATIONS & SKILLS

- **Certifications:** CompTIA Security+, AWS Solutions Architect Associate, AWS Developer Associate
- **Languages:** JavaScript, TypeScript, C#, HTML/CSS, Python, Java
- **Libraries/Frameworks:** React.js, Next.js, AngularJS, Unity Engine, .NET Core, ASP.NET Core, Entity Framework Core, Spring Boot, Redux.js, Tailwind CSS
- **Development Tools:** PostgreSQL, NoSQL, Git, Docker, Figma, AWS, Agile